

# Reducing YMCA new member registration time.

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# DEFINE

# Business Case

- The YMCA membership desk is the first point of contact for new members, making the registration process an important part of the overall customer experience.
- The current registration process takes an average of **18.22 minutes** to complete. This project aims to reduce the average registration time by **20%**, to **14.6 minutes**, by **December 31, 2026**. Based on typical daily operations, the study location completes an estimated average of **six new member registrations per day**. If the project goal is achieved, approximately **21.7 minutes of staff time could be saved each day**.
- Longer registration times can contribute to front-desk wait times, reduce staff capacity, and negatively affect the experience of new members.
- Improving the registration process is expected to reduce front desk wait times, improve the experience for new members, and allow staff to assist more members using existing resources.

Although the project requires staff time to analyze the current process and implement improvements, the expected benefits include improved staff utilization, a more consistent registration process, higher customer satisfaction, and greater operational efficiency.

# Estimated Cost-Benefit Analysis

- **Costs:** Longer front-desk wait times, lost staff capacity, member frustration, risk of a negative first impression, and potential risk to member satisfaction and retention.
- **Benefits:** Less waiting, increased staff capacity, improved registration consistency, improved member experience, improved staff utilization, and more efficient front-desk operations.
- By reducing average new-member registration time from **18.22 minutes to 14.6 minutes**, the YMCA could save approximately **21.7 minutes of staff time per day**. Assuming six new registrations per day and approximately 300 operating days per year, this represents approximately **108.6 hours of staff capacity created annually**.
- As a result, improving the registration process is expected to have a positive impact on front-desk efficiency, member experience, and staff utilization while requiring minimal direct financial investment.

# Problem & Goal Statement

- **Y Metric:** New Member Registration Time (Minutes)
- **Problem:** The current YMCA new member registration process takes an average of approximately 18.22 minutes to complete.
- **Goal:** Reduce the average new member registration time by 20%, from 18.22 minutes to 14.6 minutes, by December 31, 2026.

# SIPOC

[illegible]

# MEASURE

# Operational Definition

- New Member Registration Time **is defined as** the total time required to complete the registration process for a new YMCA member, **as measured by** recording the time in minutes from when the membership associate begins entering the new member's information into the YMCA membership system until the registration is completed, payment is processed (if applicable), and the membership is activated.
- Registration time will be measured in **minutes** using a digital stopwatch or timer. The same start point, end point, and timing method will be used for every observation to ensure consistent measurement.

# Current State – Process Map

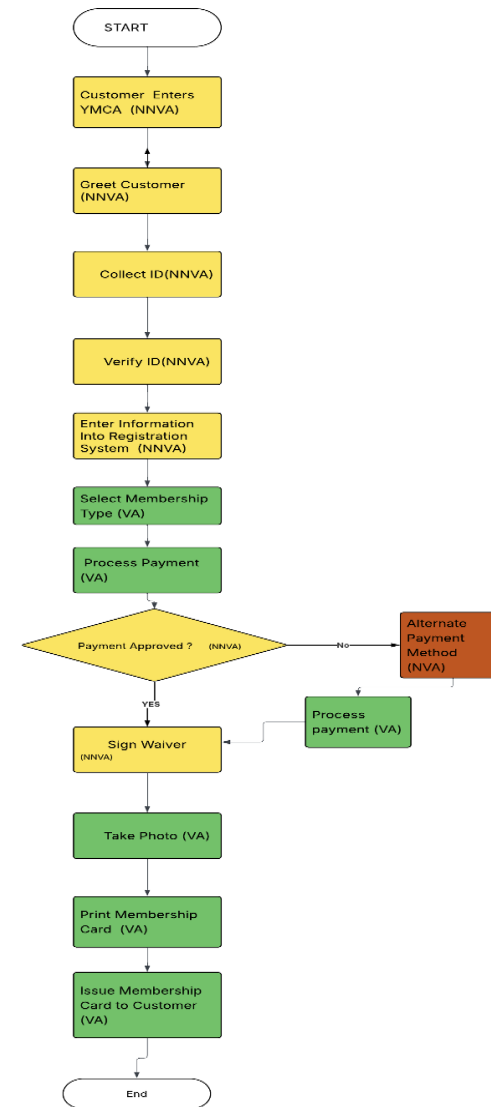
## YMCA New Member Registration Process Map

Yellow = NNVA (Necessary Non-Value Added)

Green = VA (Value Added)

Red = NVA (Non-Value Added)

## Appendix A: Data Collection



# Current State

- **Data Collection**

A dataset of **40 new member registrations** was collected across seven operating days. The Y metric, New Member Registration Time, was recorded in minutes for each observation. The data collection plan also recorded potential X variables for later analysis, including membership category, time of day, document readiness, staff experience, and payment method.

- **Current Baseline**

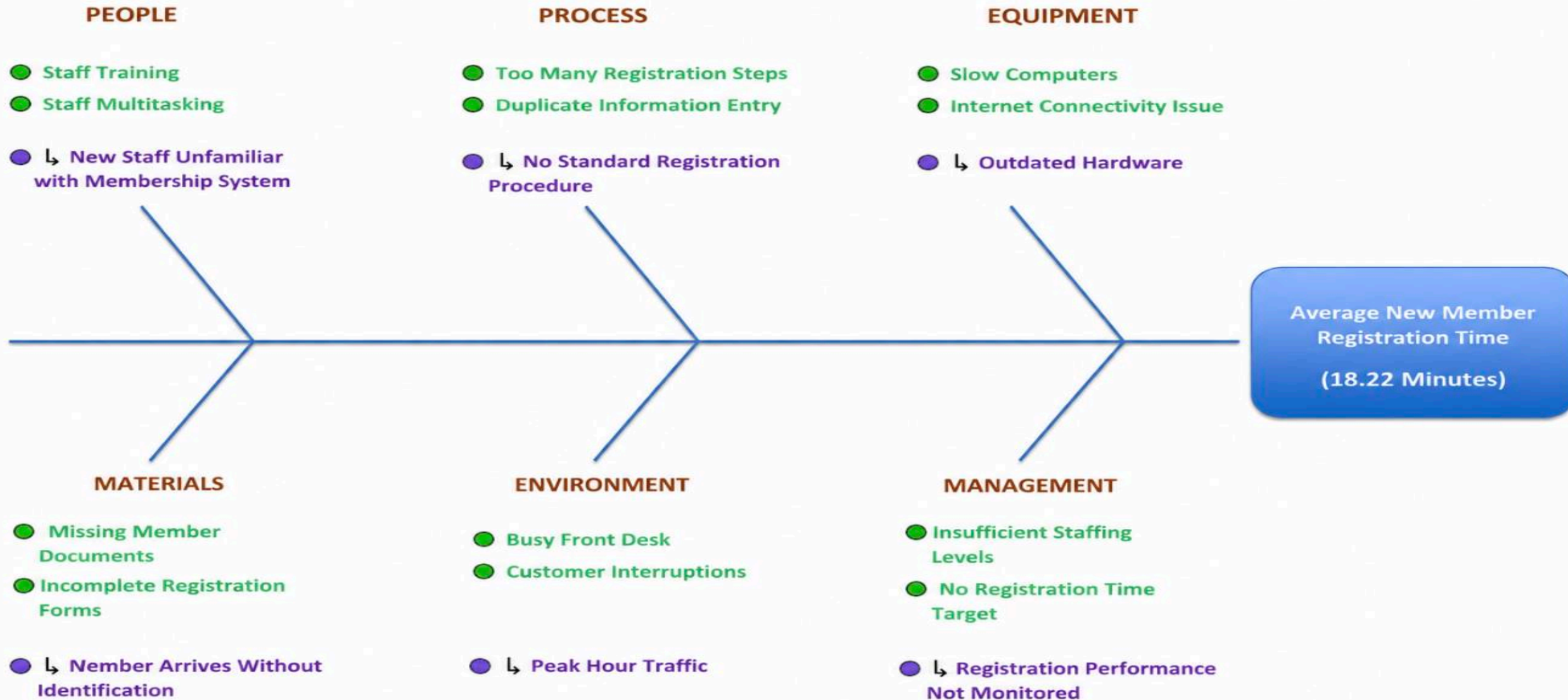
The current state of registration time will be summarized using descriptive statistics, including the sample size, mean, median, standard deviation, minimum, and maximum. A histogram and boxplot will be used to examine the distribution of registration times and determine whether the data are approximately normal so the mean will be used as the baseline measure.

- **DPMO / Sigma Level**

DPMO and Sigma Level are not applicable because New Member Registration Time is a continuous measure in minutes rather than a discrete defect measure.

# Analyze- Fishbone YMCA

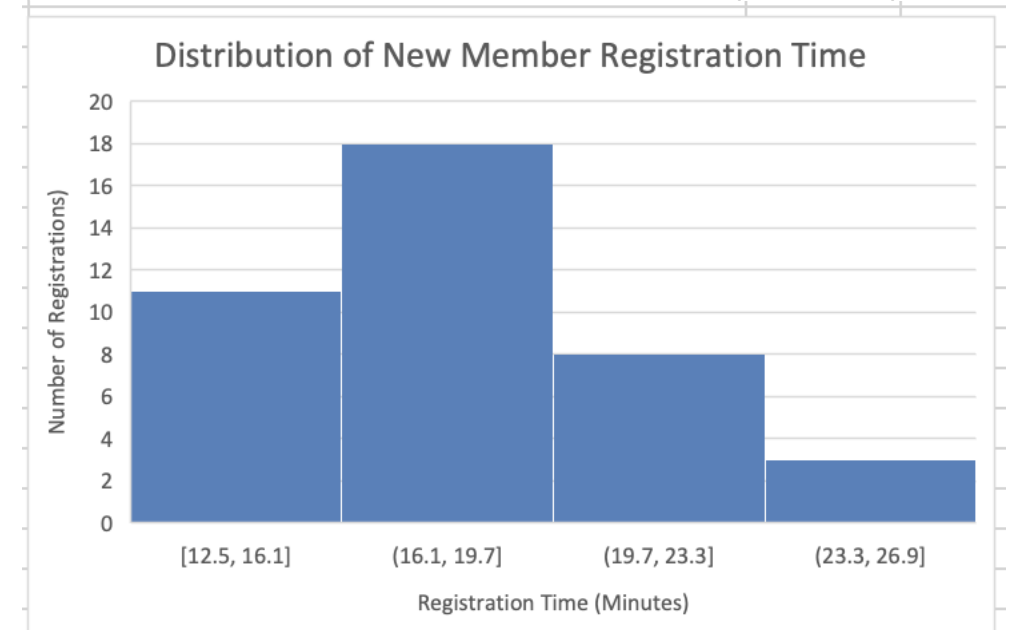
- Primary Cause
- Secondary Cause



# Current State Statistics

The histogram and boxplot show that registration times are slightly right-skewed due to a small number of longer registrations. However, the skew is limited, so the mean registration time of **18.22** minutes is used as the **current baseline**.

| Measure                              | Value |
|--------------------------------------|-------|
| Number of Registration Observations  | 40    |
| Average Registration Time (minutes)  | 18.22 |
| Median Registration Time (minutes)   | 18.65 |
| Standard Deviation(minutes)          | 3.48  |
| Shortest Registration Time (minutes) | 12.5  |
| Longest Registration Time (minutes)  | 25.8  |



# ANALYZE

# Analysis –X1: Document Readiness

## Key Findings

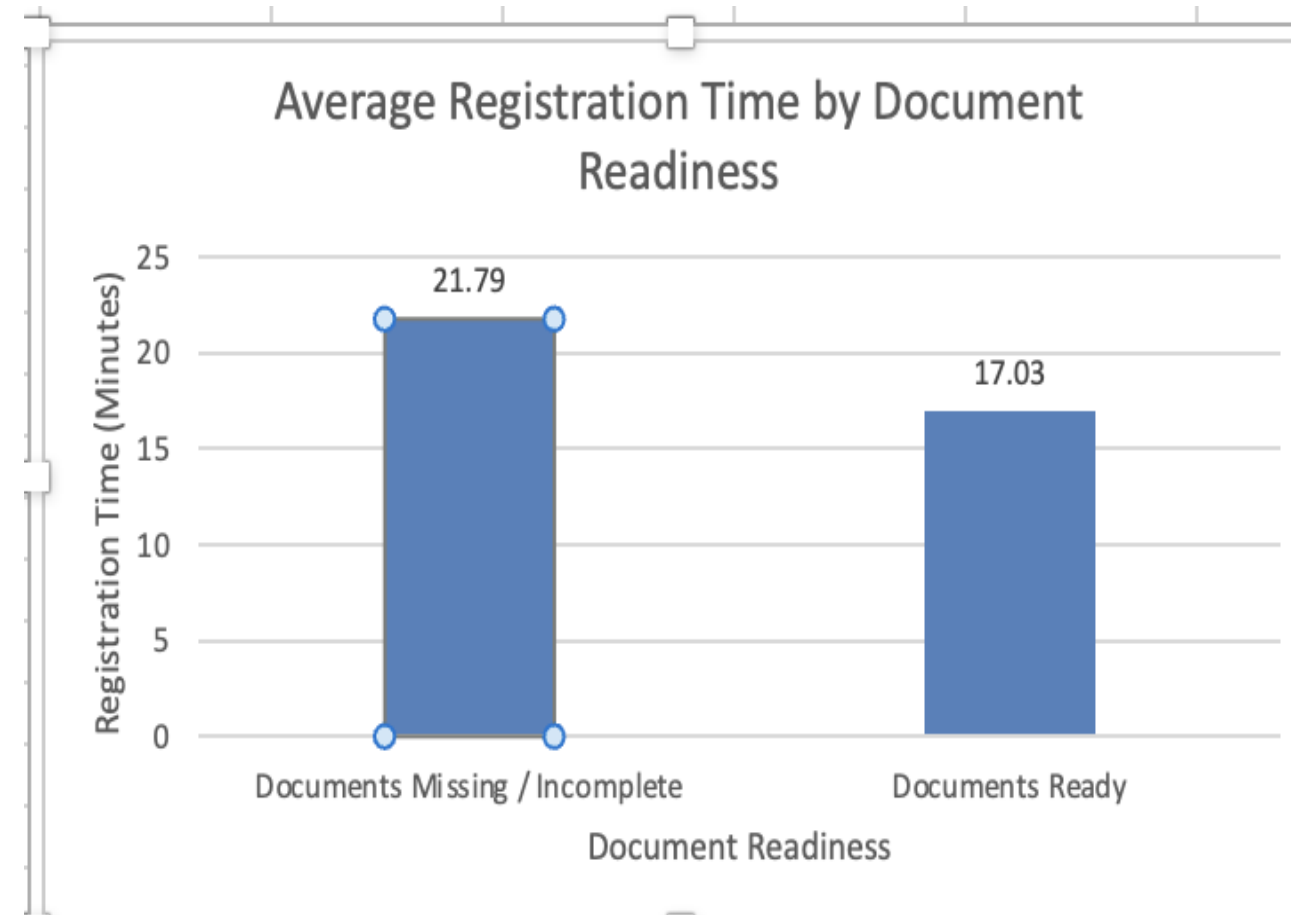
Registrations with missing or incomplete documents averaged **21.79 minutes**, compared with **17.03 minutes** when documents were ready.

Registrations with missing or incomplete documents took **4.76 minutes longer on average**.

## Statistical Test Result

A two-sample t-test found this difference to be statistically significant (**p = 0.00045**).

Document readiness appears to be a likely contributor to longer registration time.



# Analysis-X2: Staff Experience

## Key Findings

New staff averaged **20.35 minutes** per registration, compared with **17.41 minutes** for experienced staff.

Registrations completed by new staff took **2.93 minutes longer on average**.

## Statistical Test Result

A two-sample t-test found this difference to be statistically significant ( **$p = 0.00783$** ). Staff experience appears to be a likely contributor to longer registration time.

Average Registration Time by Staff Experience



# Analysis-X3: Registration Time by Time of Day

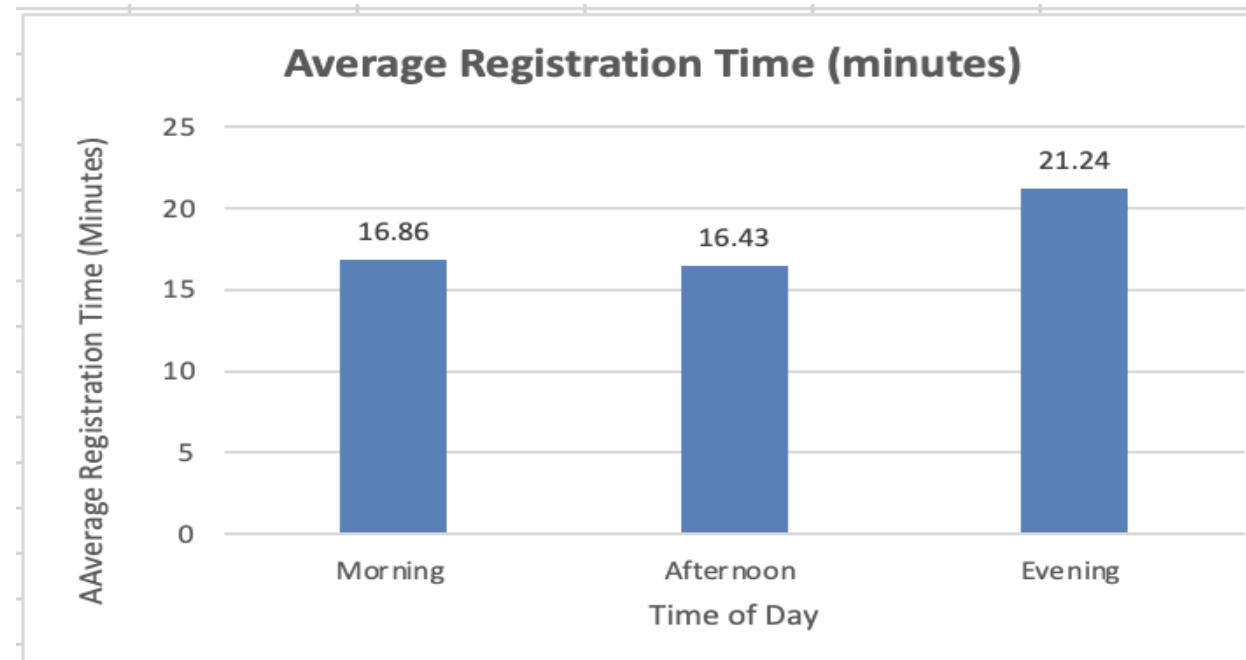
## Key Findings

Registration time varied by time of day. Morning registrations averaged **16.86 minutes**, and afternoon registrations averaged **16.43 minutes**. Evening registrations had the longest average registration time at **21.24 minutes**.

Evening registrations took approximately **4.38 minutes longer than morning registrations** and **4.82 minutes longer than afternoon registrations** on average.

## Statistical Test Result

A one-way ANOVA found a statistically significant difference in registration time across the three time-of-day groups (**p = 0.0000723**). Since the p-value is below 0.05, time of day appears to be a likely contributor to registration time. Evening registrations were associated with the longest registration times.



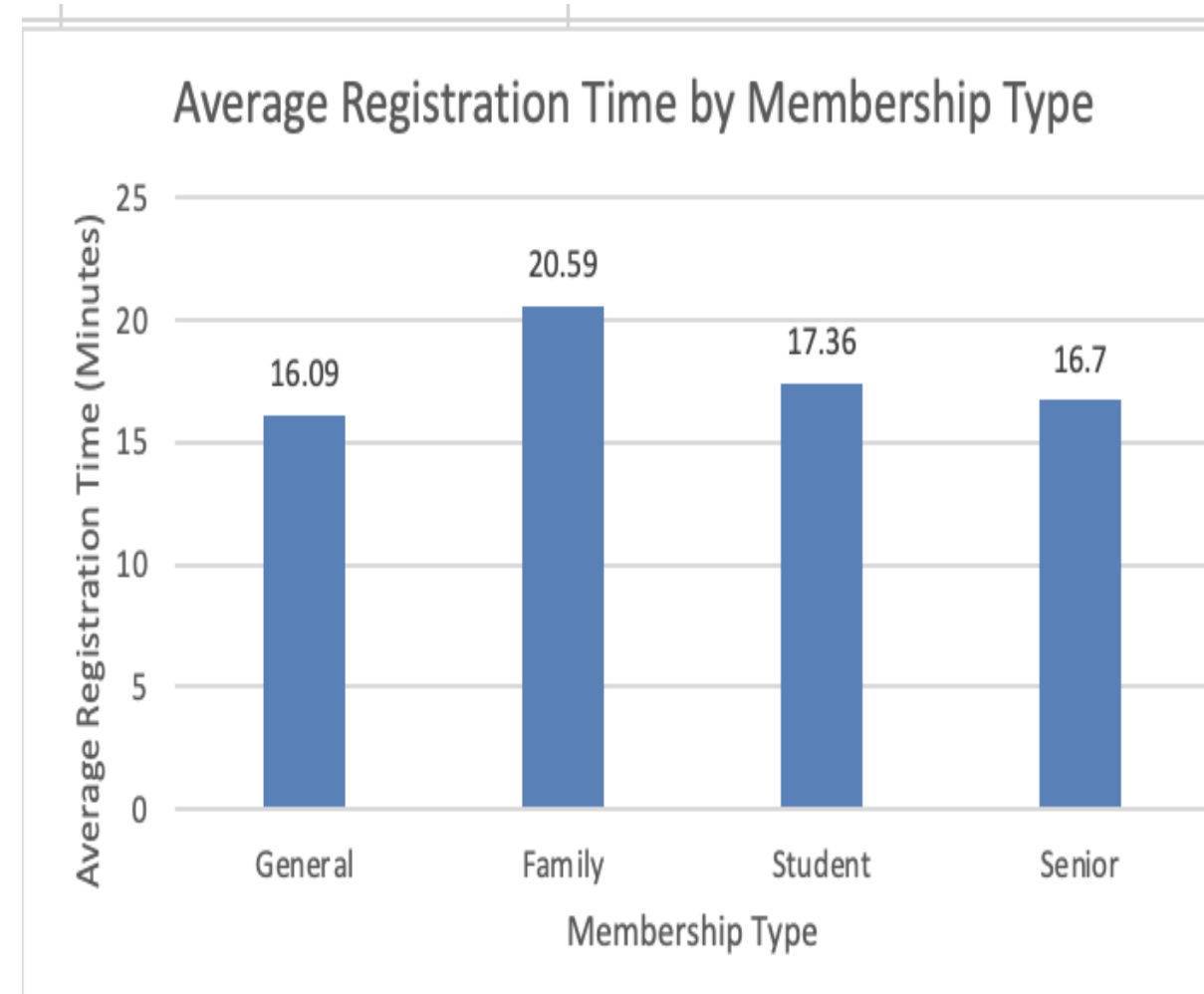
# Analysis-X4: Registration Time by Membership Type

## Key Findings

Family memberships had the longest average registration time at **20.59 minutes**. This was **4.50 minutes longer than General memberships** (16.09 minutes), **3.23 minutes longer than Student memberships** (17.36 minutes), and **3.89 minutes longer than Senior memberships** (16.70 minutes).

## Statistical Test Result

A one-way ANOVA found a statistically significant difference in registration time across membership types ( $p = 0.00209$ ). Membership type appears to be a likely contributor to registration time, with Family memberships associated with the longest average registration time.



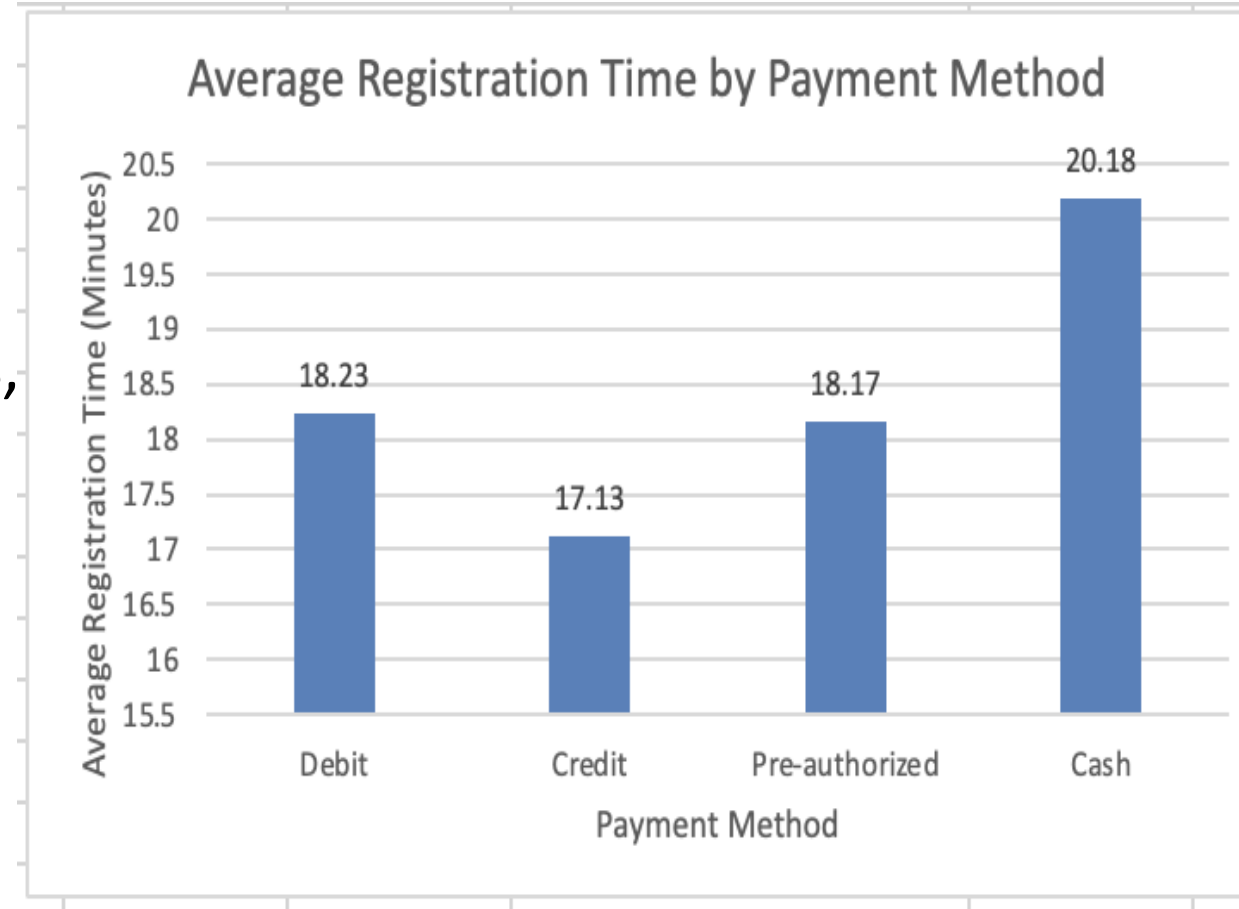
# Analysis-X5: Registration Time by Payment Method

## Key Findings

Cash registrations had the longest average registration time at **20.18 minutes**. Credit registrations had the shortest average registration time at **17.13 minutes**. Debit and pre-authorized payment registrations averaged **18.23 minutes** and **18.17 minutes**, respectively.

## Statistical Test Result

A one-way ANOVA found no statistically significant difference in registration time across payment methods ( $p = 0.48531$ ). Payment method does not appear to be a likely contributor to registration time.



# Analysis Summary: Tested X Factors

## Note:

**Two-sample t-tests** were used to compare registration time between two groups:

### **X1-Document readiness**

(ready vs. missing/incomplete) and **X2-Staff experience** (new vs. experienced).

**One-way ANOVA** was used to compare registration time across three or more groups:

**X3-Time of day** (morning, afternoon, evening) and **X4-Membership type** (General, Family, Student, Senior).

**X5 – Payment Method** (Cash, Credit, Debit, Pre-authorized).

| X factor           | Statistical Method | Visualization method   | P -value      | Effect on registration time                            | Priority       |
|--------------------|--------------------|------------------------|---------------|--------------------------------------------------------|----------------|
| Document readiness | t-test             | Clustered Column Chart | p = 0.00045   | +4.76 minutes                                          | Highest        |
| Time of day        | ANOVA              | Clustered Column Chart | p = 0.0000723 | Evening longest: 21.24 min                             | High           |
| Membership type    | ANOVA              | Clustered Column Chart | p = 0.00209   | Family longest: 20.59 min                              | High           |
| Staff experience   | t-test             | Clustered Column Chart | p = 0.00783   | +2.93 minutes                                          | Medium         |
| Payment Method     | ANOVA              | Clustered Column Chart | p = 0.48531   | Cash highest: 20.18 min; not statistically significant | Not a priority |

# RECOMMENDATIONS / CONTROL

# Recommendations

The recommendations focus on the four X factors that showed a statistically significant relationship with registration time:

**X1-Document readiness,**

**X2-staff experience,**

**X3-Time of day, and**

**X4- Membership type.**

The proposed actions are intended to reduce delays before and during registration, especially for evening and Family membership registrations.

**X5 – Payment Method** was not included because the analysis did not show a statistically significant effect on registration time.

| Validated X Factor | Recommended Action                                                                                                                                                                                       | Expected Improvement                                                                                      | Resource Need                                                                     |
|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Document readiness | Send new members a short checklist of required documents before their registration visit. At arrival, staff should confirm that required documents are available before beginning the full registration. | This addresses the 4.76 minute longer average registration time when documents are missing or incomplete. | Staff time to create the checklist and share it by email or as a printed handout. |
| Staff experience   | Create a one-page registration job aid and provide brief coaching or shadowing for new membership associates.                                                                                            | This addresses the 2.93 minute longer average registration time completed by new staff.                   | Staff time to create the job aid and provide coaching during onboarding.          |
| Time of day        | Review evening registration coverage and assign an available staff member to provide support during busy evening periods when possible.                                                                  | This addresses the longer average registration times observed during evening hours.                       | Review of the existing evening schedule, additional staffing only if needed.      |
| Membership type    | Send members registering for a family membership a short preparation guide explaining the information and documents needed for family registration.                                                      | This addresses the longer average registration time for family memberships.                               | Staff time to create and distribute the family registration guide.                |

# Control

The recommended controls require minimal direct cost because they use existing registration records, staff schedules, and the document checklists created to address

**X1 – Document Readiness,  
X2 – Staff Experience,  
X3 – Time of Day, and  
X4 – Membership Type.**

The Membership Supervisor will review the control measures regularly and take corrective action if the average registration time exceeds the 14.6-minute goal.

| Root-cause area to control | Control measure                                                                        | Frequency | Owner                 | Corrective action                                                                                                              |
|----------------------------|----------------------------------------------------------------------------------------|-----------|-----------------------|--------------------------------------------------------------------------------------------------------------------------------|
| Document readiness         | Percentage of new-member registrations beginning with all required documents available | Weekly    | Membership Supervisor | If incomplete-document cases increase, remind staff to use the checklist and update the pre-registration instructions.         |
| Staff experience           | Average registration time for new staff compared with experienced staff                | Monthly   | Membership Supervisor | If new-staff times remain higher, provide additional coaching, job-aid review, and shadowing.                                  |
| Evening registrations      | Average registration time during evening shifts                                        | Monthly   | Membership Supervisor | If evening times exceed the goal, review staffing coverage and assign support during peak periods.                             |
| Family memberships         | Average registration time for Family memberships                                       | Monthly   | Membership Supervisor | If Family registrations remain longer, revise the family preparation guide and identify commonly missing information.          |
| Overall project goal       | Average new-member registration time                                                   | Weekly    | Membership Supervisor | If the weekly average exceeds <b>14.4 minutes</b> , review delayed registrations and identify which X factor caused the delay. |

# APPENDIX

# Appendix B : X1-Document Readiness t-Test Results

| t-Test: Two-Sample Assuming Unequal Variances |             |             |
|-----------------------------------------------|-------------|-------------|
|                                               |             |             |
|                                               | Variable 1  | Variable 2  |
| Mean                                          | 21.79       | 17.03       |
| Variance                                      | 8.732111111 | 7.749758621 |
| Observations                                  | 10          | 30          |
| Hypothesized Mean Difference                  | 0           |             |
| df                                            | 15          |             |
| t Stat                                        | 4.474790304 |             |
| P(T<=t) one-tail                              | 0.000222562 |             |
| t Critical one-tail                           | 1.753050356 |             |
| P(T<=t) two-tail                              | 0.000445123 |             |
| t Critical two-tail                           | 2.131449546 |             |

# Appendix C: X2- Staff Experience t-Test Results

| A                                             | B                 | C                 |
|-----------------------------------------------|-------------------|-------------------|
| t-Test: Two-Sample Assuming Unequal Variances |                   |                   |
|                                               |                   |                   |
|                                               | <i>Variable 1</i> | <i>Variable 2</i> |
| Mean                                          | 17.4137931        | 20.34545455       |
| Variance                                      | 12.08051724       | 6.646727273       |
| Observations                                  | 29                | 11                |
| Hypothesized Mean Difference                  | 0                 |                   |
| df                                            | 24                |                   |
| t Stat                                        | -2.901614836      |                   |
| P(T<=t) one-tail                              | 0.003915199       |                   |
| t Critical one-tail                           | 1.71088208        |                   |
| P(T<=t) two-tail                              | 0.007830398       |                   |
| t Critical two-tail                           | 2.063898562       |                   |

# Appendix D: X3 –Time of Day ANOVA Results

| A                        | B            | C          | D              | E               | F              |    |
|--------------------------|--------------|------------|----------------|-----------------|----------------|----|
| Anova: Single Factor     |              |            |                |                 |                |    |
| SUMMARY                  |              |            |                |                 |                |    |
| <i>Groups</i>            | <i>Count</i> | <i>Sum</i> | <i>Average</i> | <i>Variance</i> |                |    |
| Morning                  | 12           | 202.3      | 16.85833333    | 6.84265152      |                |    |
| Afternoon                | 12           | 197.1      | 16.425         | 7.97477273      |                |    |
| Evening                  | 14           | 297.4      | 21.2428571     | 7.4610989       |                |    |
| ANOVA                    |              |            |                |                 |                |    |
| <i>Source of Variati</i> | <i>SS</i>    | <i>df</i>  | <i>MS</i>      | <i>F</i>        | <i>P-value</i> |    |
| Between Grou             | 188.322469   | 2          | 94.1612343     | 12.6762357      | 7.2282E-05     | 3. |
| Within Group             | 259.985952   | 35         | 7.42817007     |                 |                |    |

# Appendix E: X4- Membership Type ANOVA Results

| A                    | B          | C     | D          | E          | F          | G          |
|----------------------|------------|-------|------------|------------|------------|------------|
| Anova: Single Factor |            |       |            |            |            |            |
| SUMMARY              |            |       |            |            |            |            |
| Groups               | Count      | Sum   | Average    | Variance   |            |            |
| General              | 11         | 177   | 16.0909091 | 8.81690909 |            |            |
| Family               | 16         | 329.4 | 20.5875    | 6.61983333 |            |            |
| Student              | 8          | 138.9 | 17.3625    | 15.5941071 |            |            |
| Senior               | 5          | 83.5  | 16.7       | 4.915      |            |            |
| ANOVA                |            |       |            |            |            |            |
| Source of Variation  | SS         | df    | MS         | F          | P-value    | F crit     |
| Between Groups       | 156.978659 | 3     | 52.3262197 | 5.95583692 | 0.00209107 | 2.86626555 |
| Within Groups        | 316.285341 | 36    | 8.78570391 |            |            |            |
| Total                | 473.264    | 39    |            |            |            |            |

## Appendix F: X5 – Payment Method ANOVA Results

| Anova: Single Factor       |              |            |                |                 |                |               |
|----------------------------|--------------|------------|----------------|-----------------|----------------|---------------|
|                            |              |            |                |                 |                |               |
| SUMMARY                    |              |            |                |                 |                |               |
| <i>Groups</i>              | <i>Count</i> | <i>Sum</i> | <i>Average</i> | <i>Variance</i> |                |               |
| Debit                      | 19           | 346.4      | 18.2315789     | 11.2733918      |                |               |
| Credit                     | 10           | 171.3      | 17.13          | 7.88455556      |                |               |
| Pre-authorized             | 7            | 127.2      | 18.1714286     | 15.7357143      |                |               |
| Cash                       | 4            | 80.7       | 20.175         | 6.7825          |                |               |
|                            |              |            |                |                 |                |               |
|                            |              |            |                |                 |                |               |
| ANOVA                      |              |            |                |                 |                |               |
| <i>Source of Variation</i> | <i>SS</i>    | <i>df</i>  | <i>MS</i>      | <i>F</i>        | <i>P-value</i> | <i>F crit</i> |
| Between Groups             | 26.9321617   | 3          | 8.97738722     | 0.83157356      | 0.48531107     | 2.86626555    |
| Within Groups              | 388.643838   | 36         | 10.7956622     |                 |                |               |
|                            |              |            |                |                 |                |               |
| Total                      | 415.576      | 39         |                |                 |                |               |